Name
Date
Course

Practice Geometry Test III (Version I)

Instructions

I) Write your name, the date, and the course title (i.e. Geometry).
II) Show work when necessary.
III) If a problem requires algebraic steps to solve, draw a rectangle around your final answer to distinguish it from the other work.
IV) Each problem is worth four points for a total of 100 points.
V) A calculator may be useful for some problems on this test.

① If the pair of numbers below represents side lengths of a triangle, write an inequality that shows the possible lengths of the third side X.

48, 31

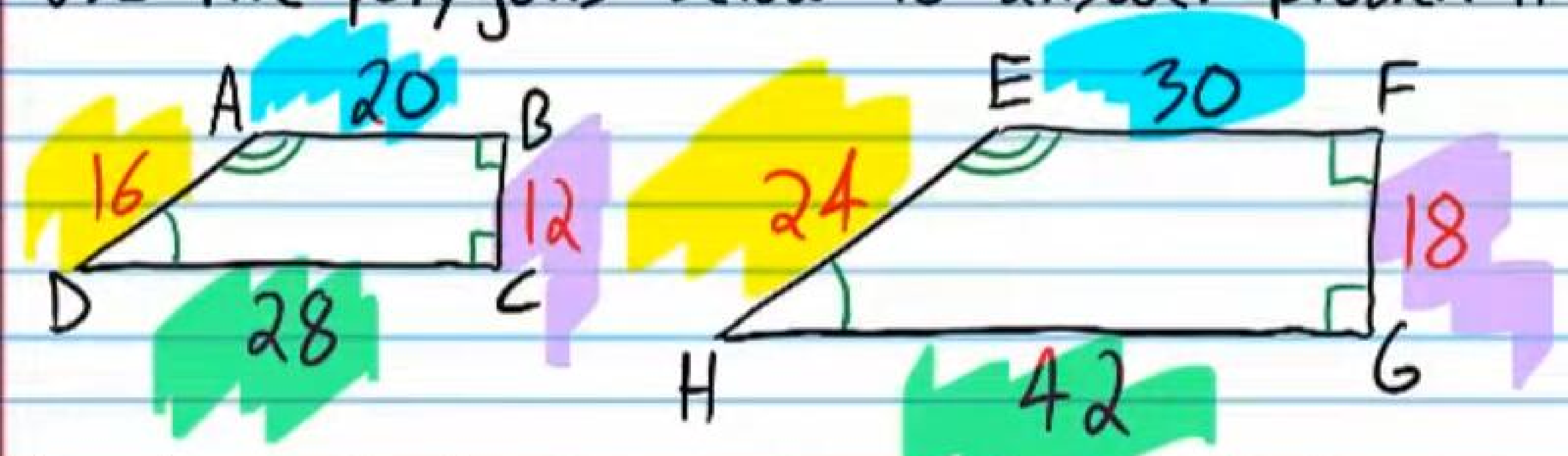
X, 48, 31

a) $X+48>31$ b) $X+31>48$ c) $48+31>X$

$X>-17$ $X>17$ $79>X$

$$\boxed{17<X<79}$$

Use the polygons below to answer problem #'s 2-4.



② Show that the polygons are similar.

a) $\boxed{\angle A \cong \angle E, \angle B \cong \angle F, \angle D \cong \angle H, \& \angle C \cong \angle G}$

b) $\frac{16}{24} = \frac{20}{30} = \frac{12}{18} = \frac{28}{42}$

$\frac{2}{3} = \frac{2}{3} = \frac{2}{3} = \frac{2}{3}$

$$\boxed{\frac{AD}{HE} = \frac{AB}{EF} = \frac{BC}{FG} = \frac{DC}{HG}}$$

③ Write a similarity statement: $\boxed{\text{▱}ABCD \sim \text{▱}EFGH}$

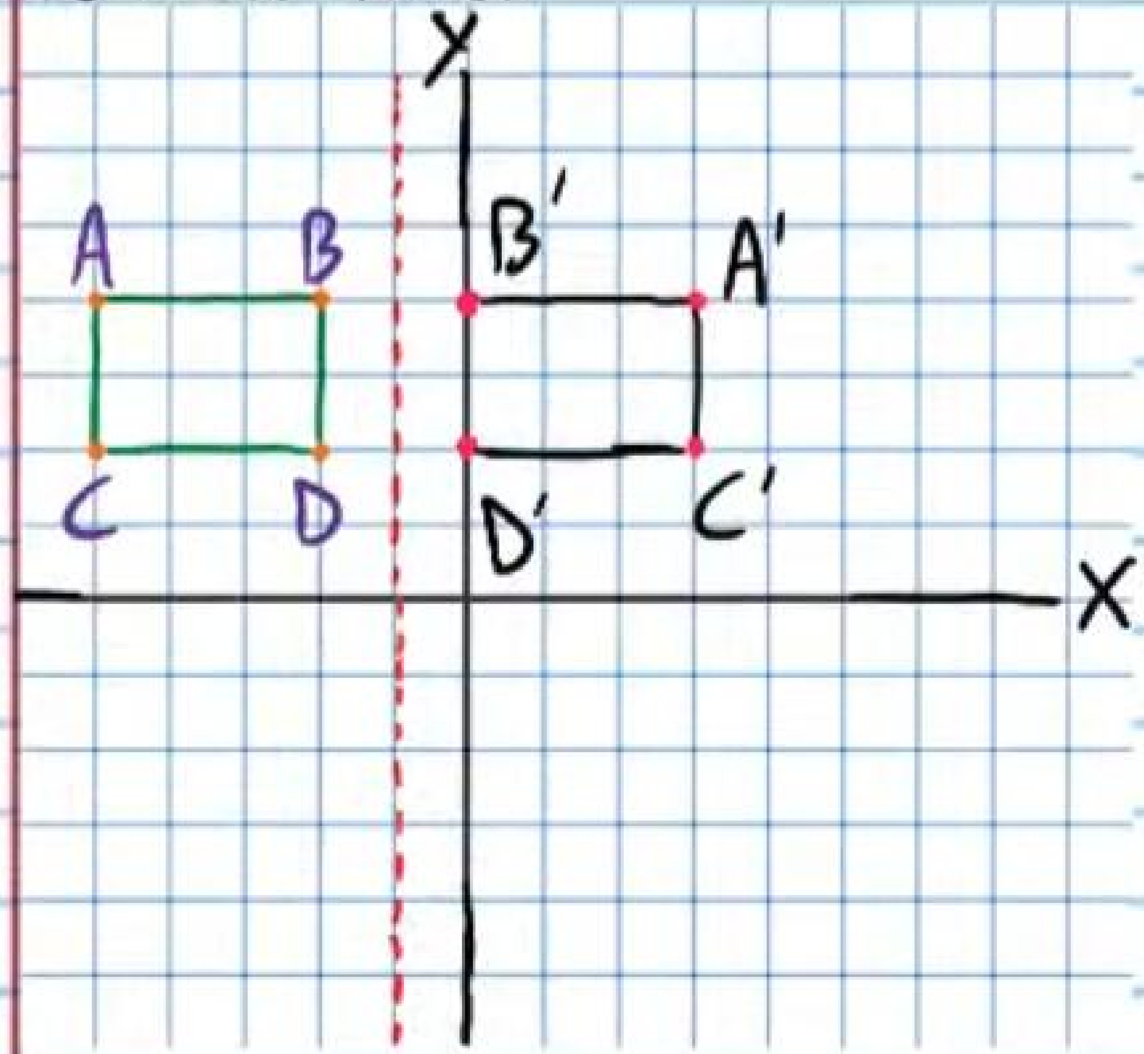
④ i) Write the scale factor: $\boxed{\frac{2}{3}}$

ii) Write the scale factor in a different way (Hint: Remember, there are two ways to write a scale factor).

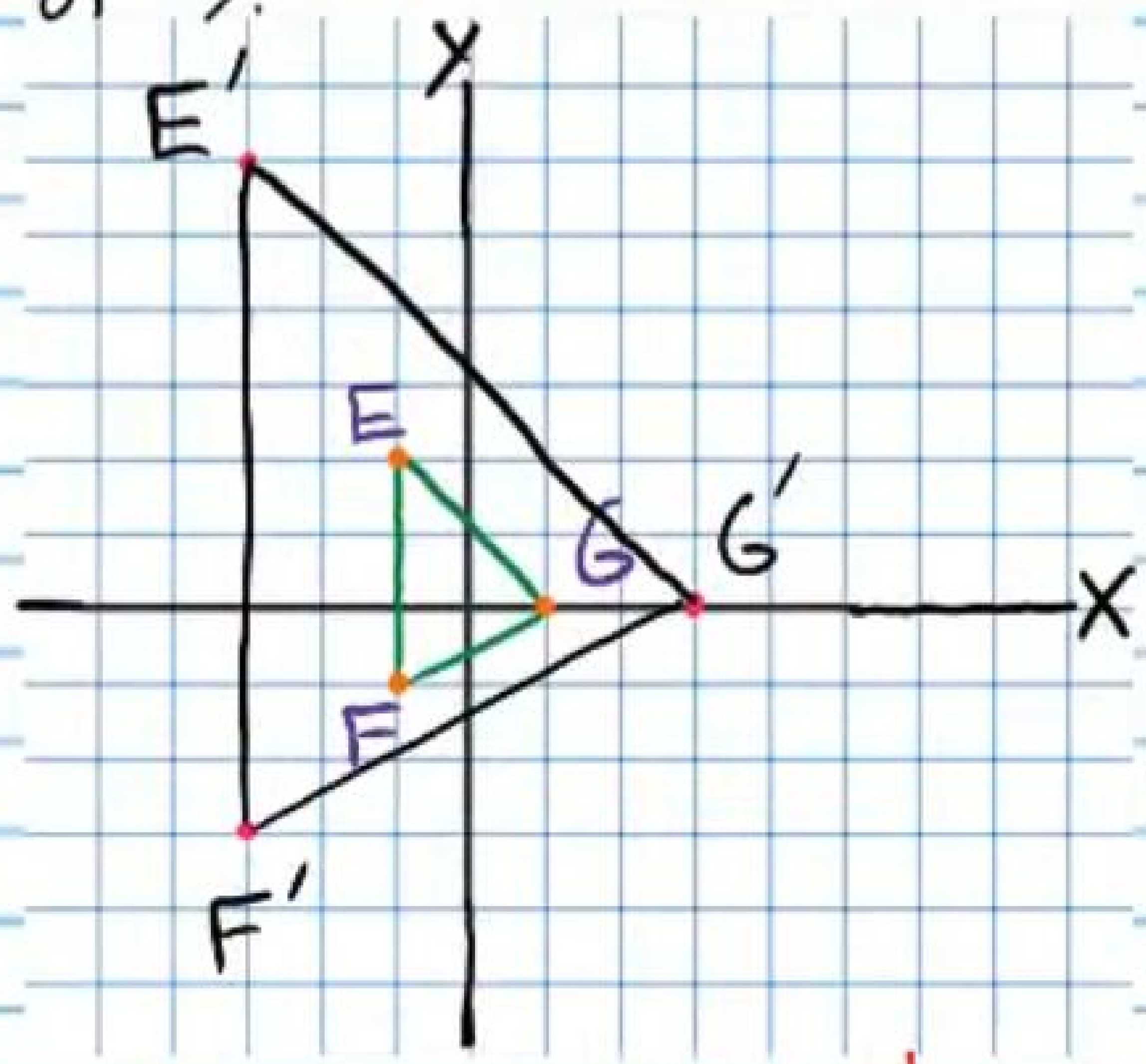
$$\boxed{\frac{3}{2}}$$

Given the "pre-image," graph the "image" using the given transformation.

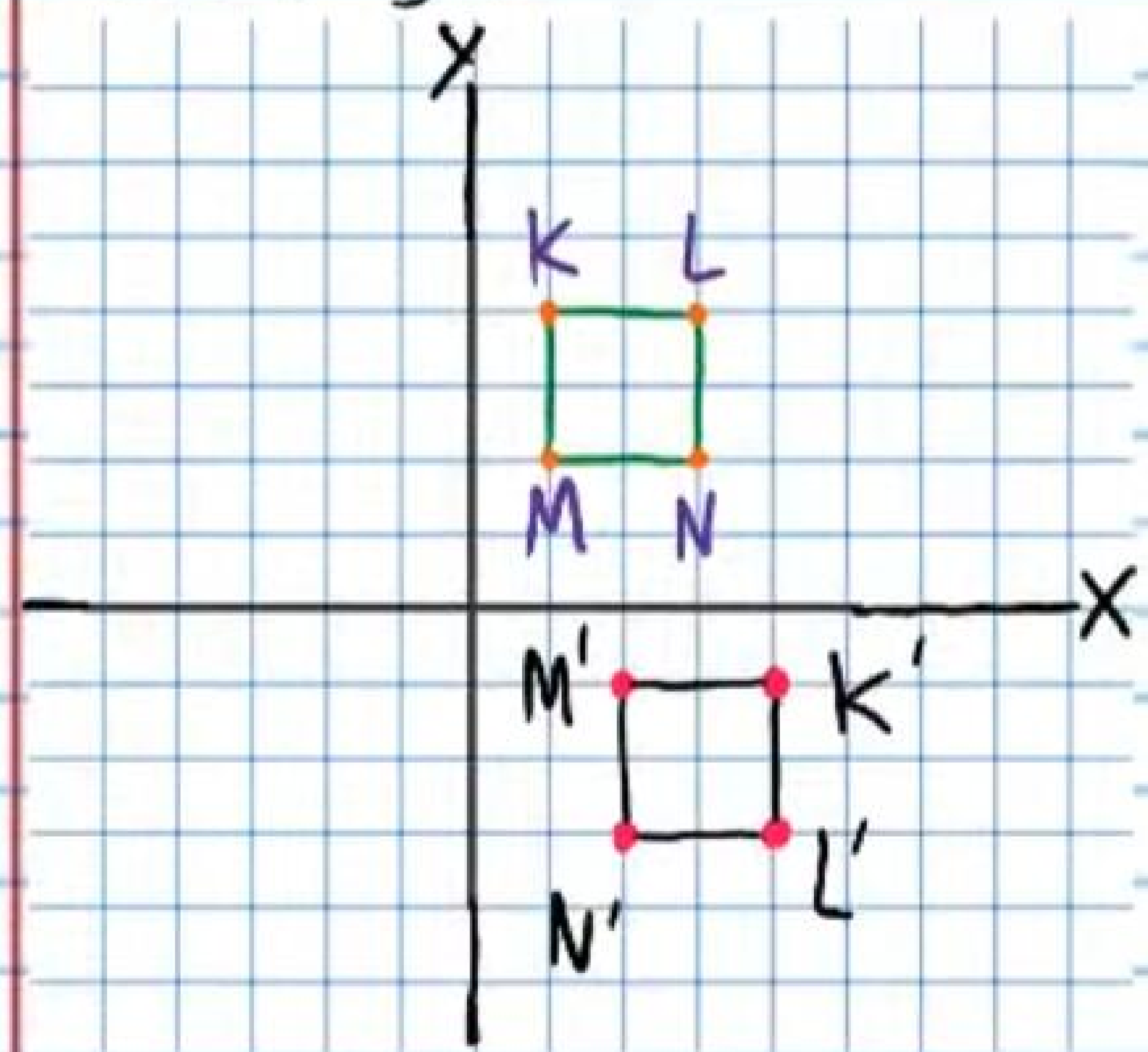
⑤ Reflection about the dotted line.



⑥ Dilation about the origin with a scale factor of 3.



⑦ Rotation: $-90°$ about the origin.



⑧ Is it possible to make a triangle with the group of side lengths below? If not, write the inequality that shows that it is not possible.

8, 12, 4

a) $8+12>4$

$20>4$

✓

b) $\boxed{8+4 \not> 12}$

$12 \not> 12$

✗

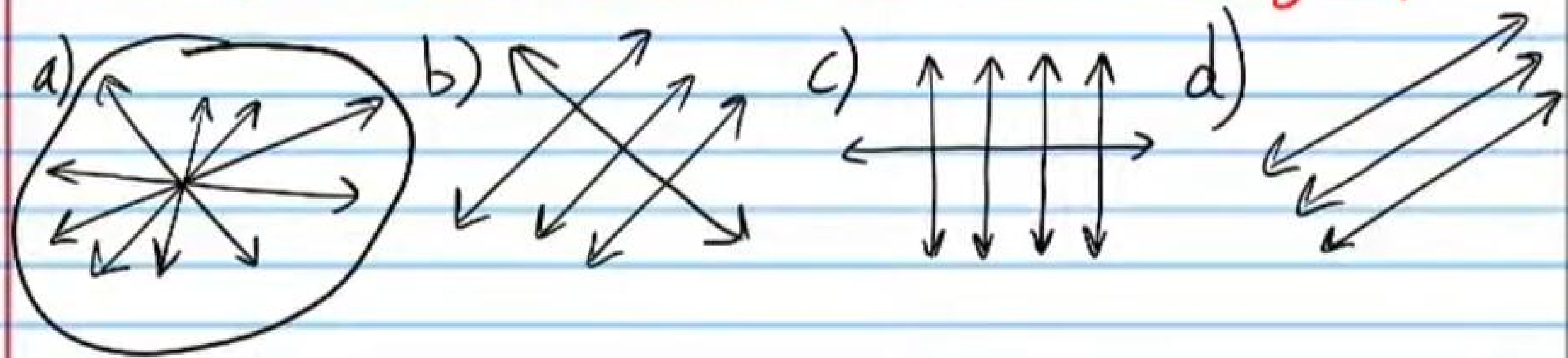
No

c) $12+4>8$

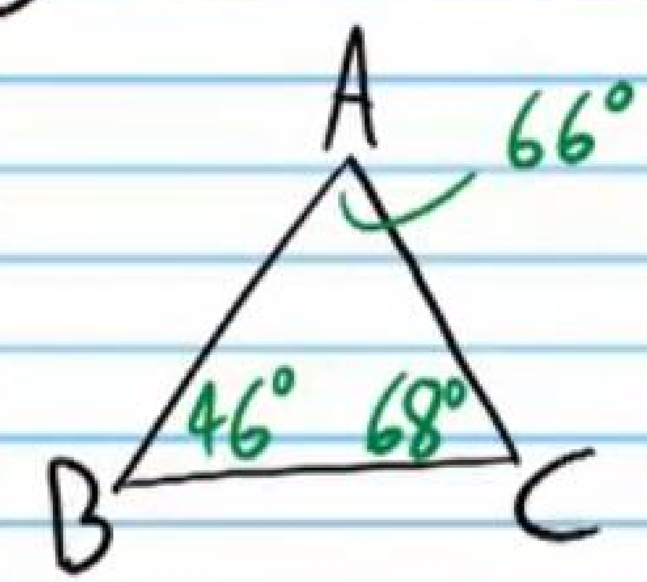
$16>8$

✓

⑨ Circle each group that contains concurrent lines.
You may need to circle more than one group.

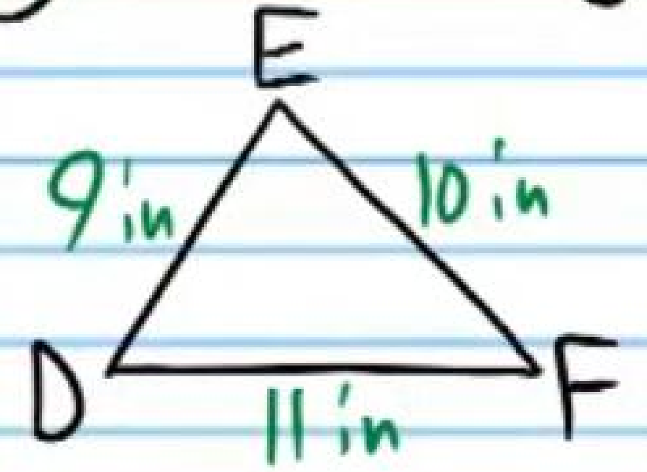


(10) List the sides of the triangle from shortest to longest.



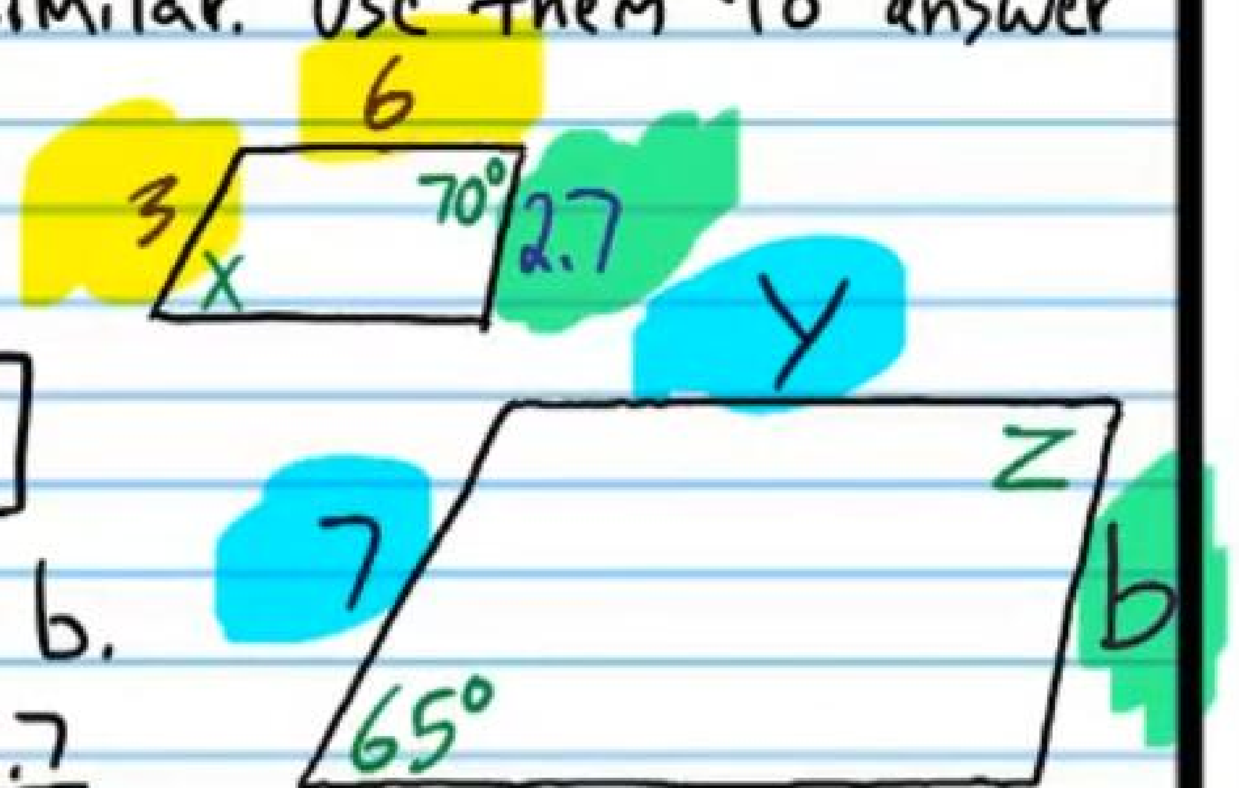
$\boxed{AC, BC, AB}$

(11) List the angles of the triangle from least to greatest.



$\boxed{\angle F, \angle D, \angle E}$

The polygons below are similar. Use them to answer problem #'s 12-13.



(12) i) Find X. ii) Find Z.

$\boxed{X = 65°}$ $\boxed{Z = 70°}$

(13) i) Find y.

$$\frac{3}{7} = \frac{6}{y}$$

$$\frac{(y)\,3}{(y)\,7} = \frac{6\,(7)}{y\,(7)}$$

$$\frac{3y}{7y} = \frac{42}{7y}$$

$$3y = 42$$

$$\boxed{y = 14}$$

ii) Find b.

$$\frac{3}{7} = \frac{2.7}{b}$$

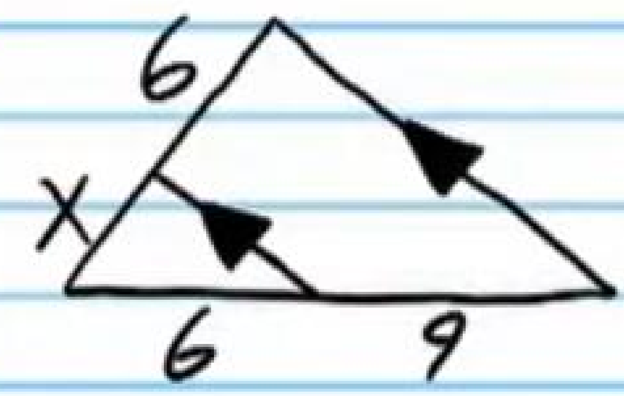
$$\frac{b \cdot 3}{b \cdot 7} = \frac{2.7 \cdot 7}{b \cdot 7}$$

$$\frac{3b}{7b} = \frac{18.9}{7b}$$

$$3b = 18.9$$

$$\boxed{b = 6.3}$$

(14) Find X.



$$\frac{6}{x} = \frac{9}{6}$$

$$\frac{6}{x} = \frac{3}{2}$$

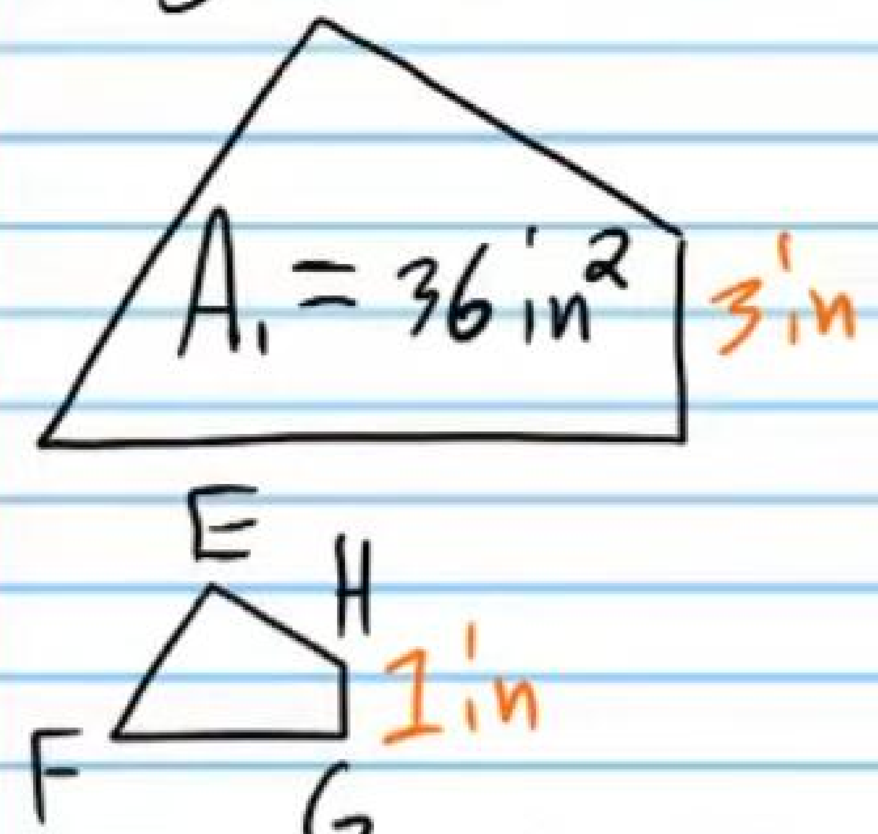
$$\frac{2 \cdot 6}{2 \cdot x} = \frac{3 \cdot x}{2 \cdot x}$$

$$\frac{12}{2x} = \frac{3x}{2x}$$

$$12 = 3x$$

$$\boxed{4 = x}$$

(15) Find the area of EFGH if the two polygons below are similar. Show your proportion.



$A_2 = ?$

$$\frac{A_1}{A_2} = \left(\frac{3}{1}\right)^2$$

$$\frac{36}{A_2} = 3^2$$

$$\frac{36}{A_2} = 9$$

$$\frac{36}{A_2} = \frac{9}{1}$$

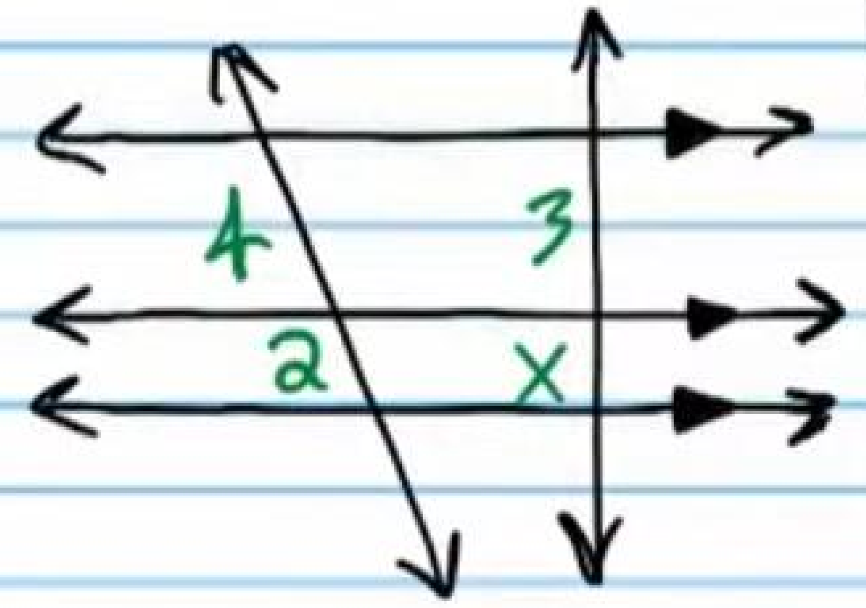
$$\frac{36}{A_2} = \frac{9 \cdot A_2}{1 \cdot A_2}$$

$$\frac{36}{A_2} = \frac{9A_2}{A_2}$$

$$36 = 9A_2$$

$$\boxed{4\,in^2 = A_2}$$

(16) Find x.



$$\frac{4}{2}=\frac{3}{x}$$

$$\frac{2}{1}=\frac{3}{x}$$

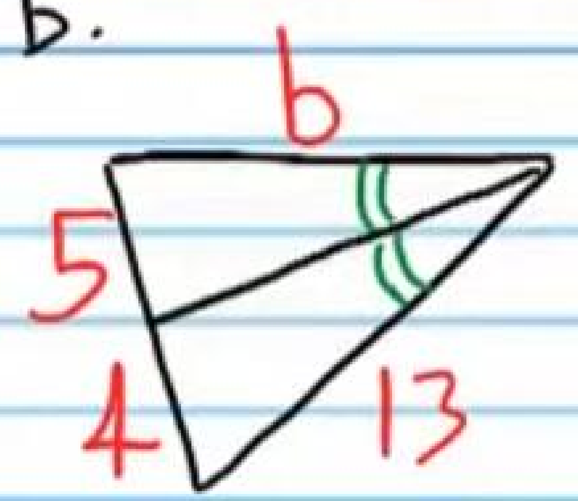
$$\frac{x\cdot 2}{x\cdot 1}=\frac{3}{x}$$

$$\frac{2x}{\cancel{x}}=\frac{3}{\cancel{x}}$$

$$2x=3$$

$$\boxed{x=\frac{3}{2}}$$

(17) Find b.



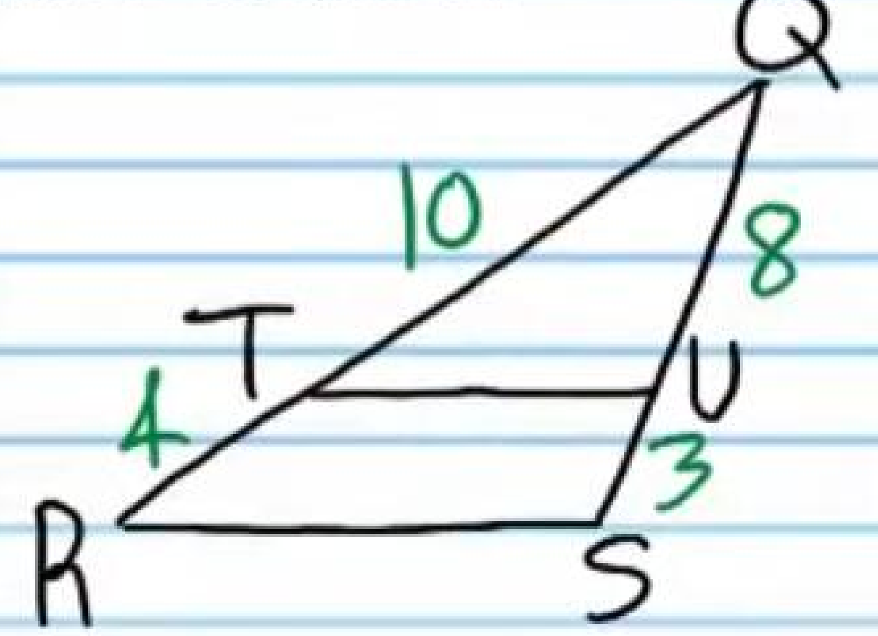
$$\frac{5}{4}=\frac{b}{13}$$

$$13\cdot\frac{5}{4}=\frac{b}{13}\cdot 13$$

$$\frac{13}{1}\cdot\frac{5}{4}=b$$

$$\boxed{\frac{65}{4}=b}$$

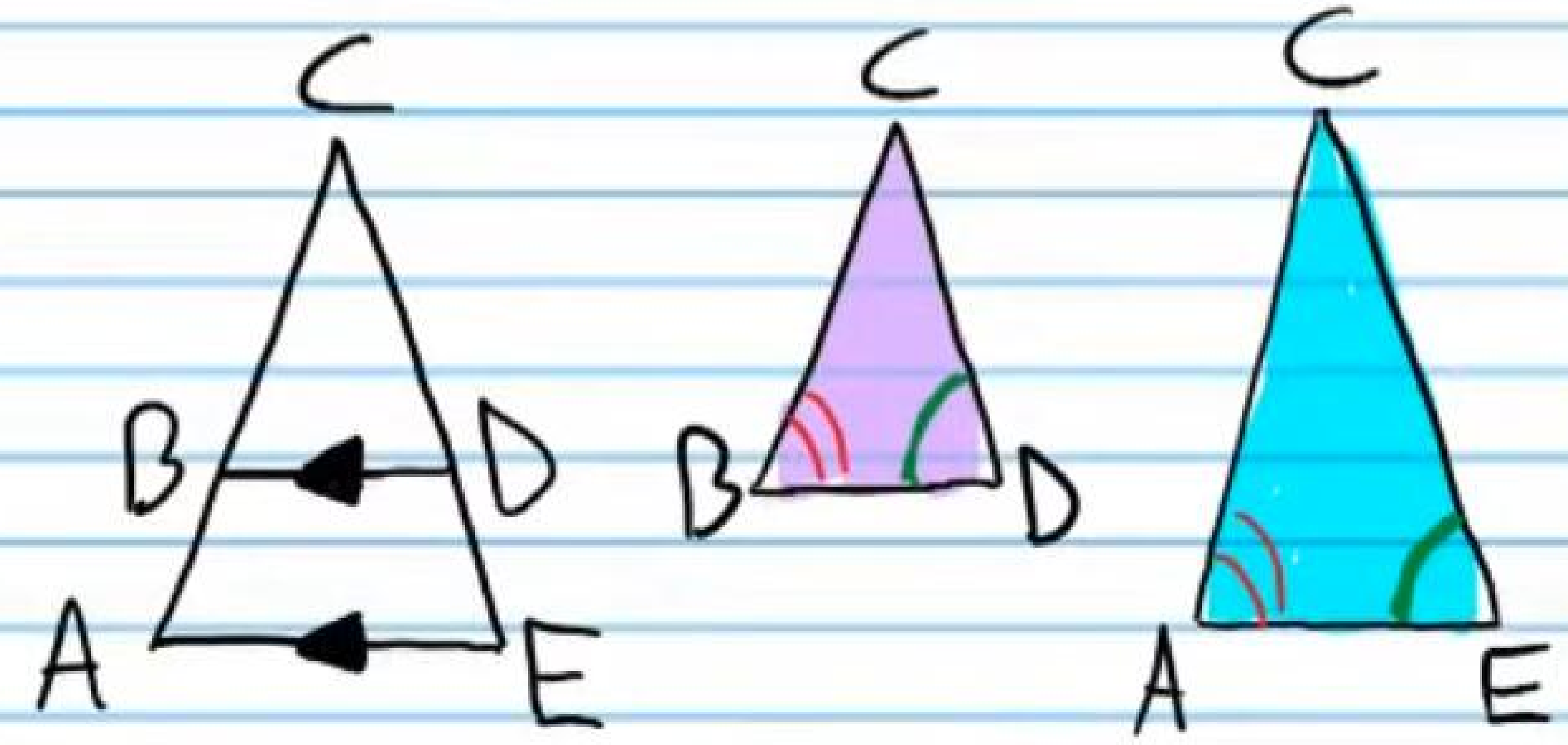
(18) Is $\overline{TU} \parallel \overline{RS}$? Show the proportions that prove your answer.



$$\boxed{\begin{array}{c}\frac{10}{4}\neq\frac{8}{3}\\ \frac{5}{2}\neq\frac{8}{3}\\ \text{NO}\end{array}}$$

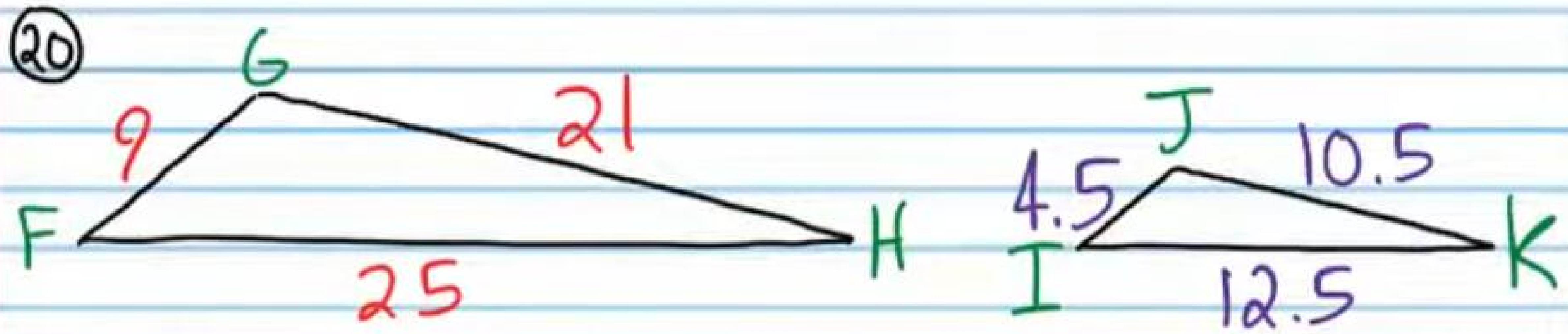
In the following problems, write two-column proofs showing that the triangles are similar. You might not use the same number of steps written in Roman numerals. You may use more or less.

(19)



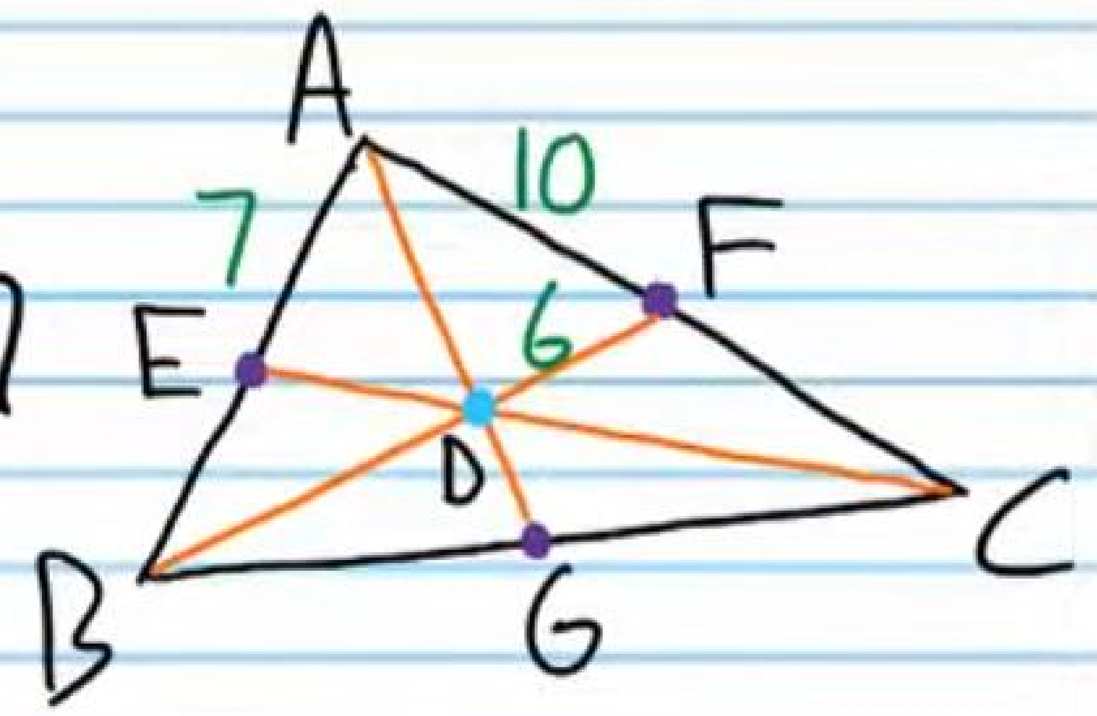
Statement	Reason
I) $\overline{BD} \parallel \overline{AE}$	Given (diagram)
II) $\angle BDC \cong \angle E$ & $\angle CBD \cong \angle A$	CAP
III) $\triangle CBD \sim \triangle CAE$	AA

⑳



Statement	Reason
I) $FG=9$, $GH=21$, $FH=25$, $IJ=4.5$, $JK=10.5$, & $IK=12.5$	Given (diagram)
II) $\frac{FG}{IJ}=\frac{9}{4.5}$, $\frac{GH}{JK}=\frac{21}{10.5}$, & $\frac{FH}{IK}=\frac{25}{12.5}$	Subs. Prop of =.
III) $\frac{FG}{IJ}=2$, $\frac{GH}{JK}=2$, & $\frac{FH}{IK}=2$	Simplification
IV) $\frac{FG}{IJ}=\frac{GH}{JK}=\frac{FH}{IK}$	Trans. Prop of =.
V) $\triangle FGH \sim \triangle IJK$	SSS

㉑ If the segments below are the medians of the triangle, find the quantities listed.



i) $BE = \boxed{7}$

ii) $BD = 2FE = 2(6) = \boxed{12}$

iii) $AC = 10 + FC$

$= 10 + 10$

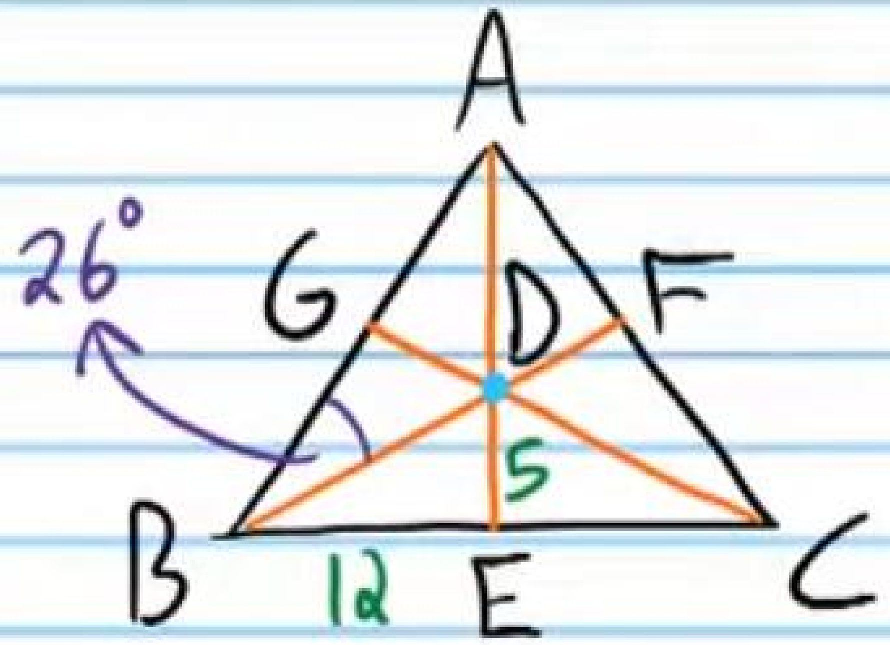
$= \boxed{20}$

iv) If $ED = 7$, find DC.

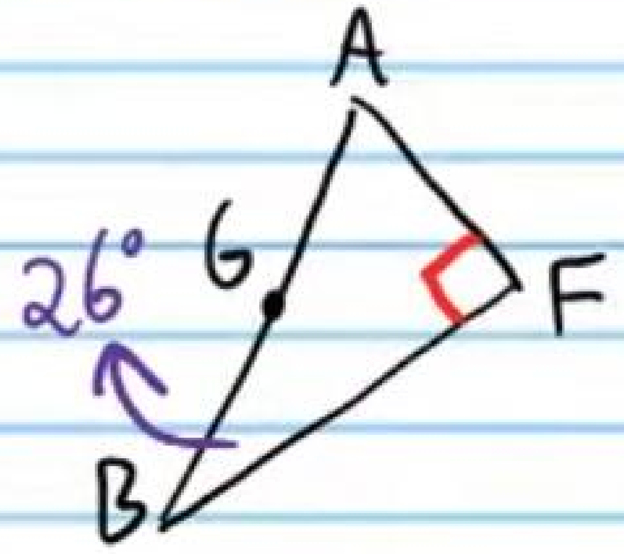
$DC = 2ED = 2(7) = \boxed{14}$

(22) If point D is the orthocenter of the triangle below, find the quantities listed.

i) $m\angle BEA = 90°$



ii) $m\angle GAF$

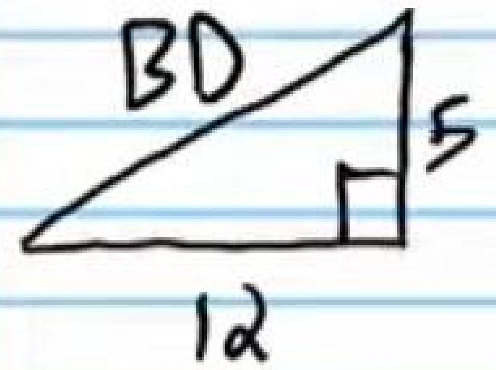


$26 + 90 + m\angle GAF = 180$

$116 + m\angle GAF = 180$

$\boxed{m\angle GAF = 64°}$

iii) BD



$5^2 + 12^2 = BD^2$

$25 + 144 = BD^2$

$169 = BD^2$

$\pm\sqrt{169} = \sqrt{BD^2}$

$\pm 13 = BD$

$\boxed{13 = BD}$

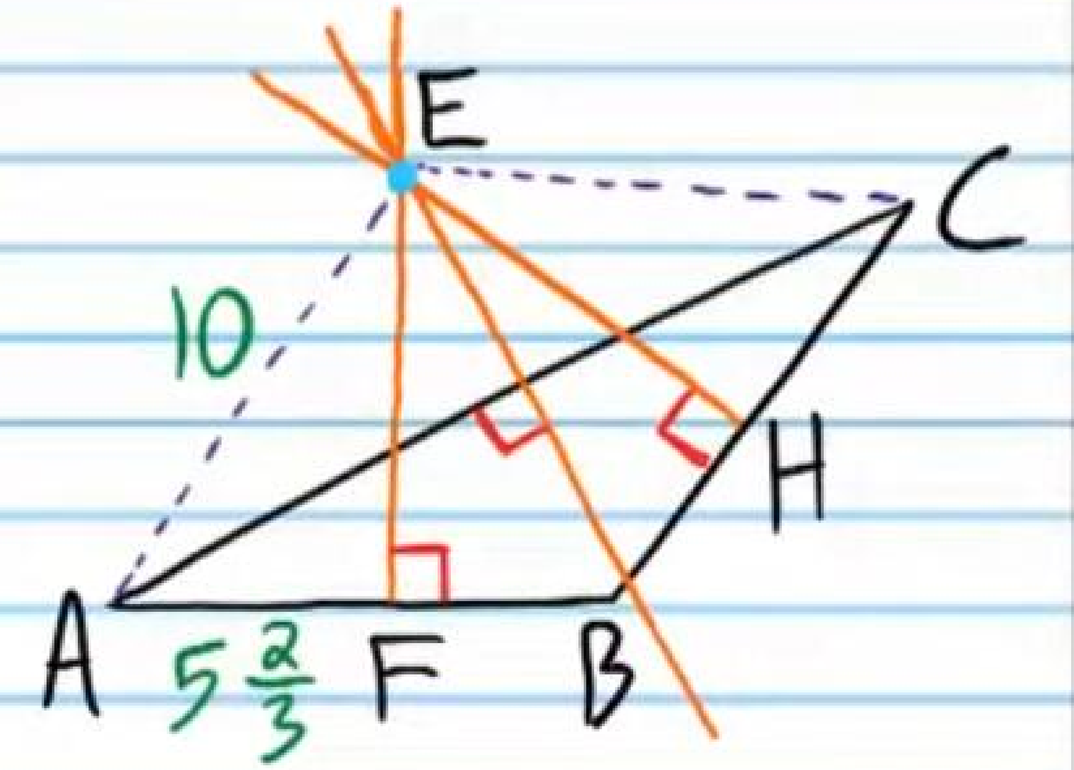
iv) $m\angle AGC = \boxed{90°}$

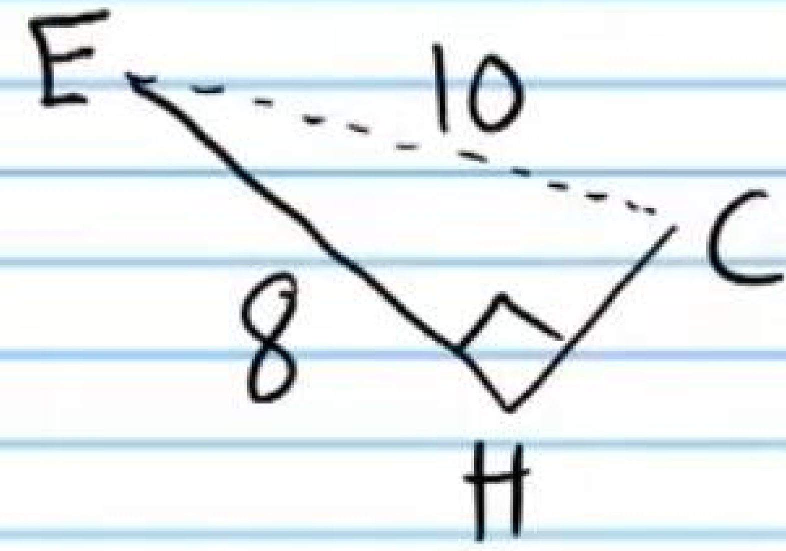
(23) If point E is the circumcenter of the triangle below, find the quantities listed.

i) $FB = \boxed{5\frac{2}{3}}$

ii) $EC = \boxed{10}$

iii) If $HE = 8$, find CH.





$8^2 + HC^2 = 10^2$

$64 + HC^2 = 100$

$HC^2 = 36$

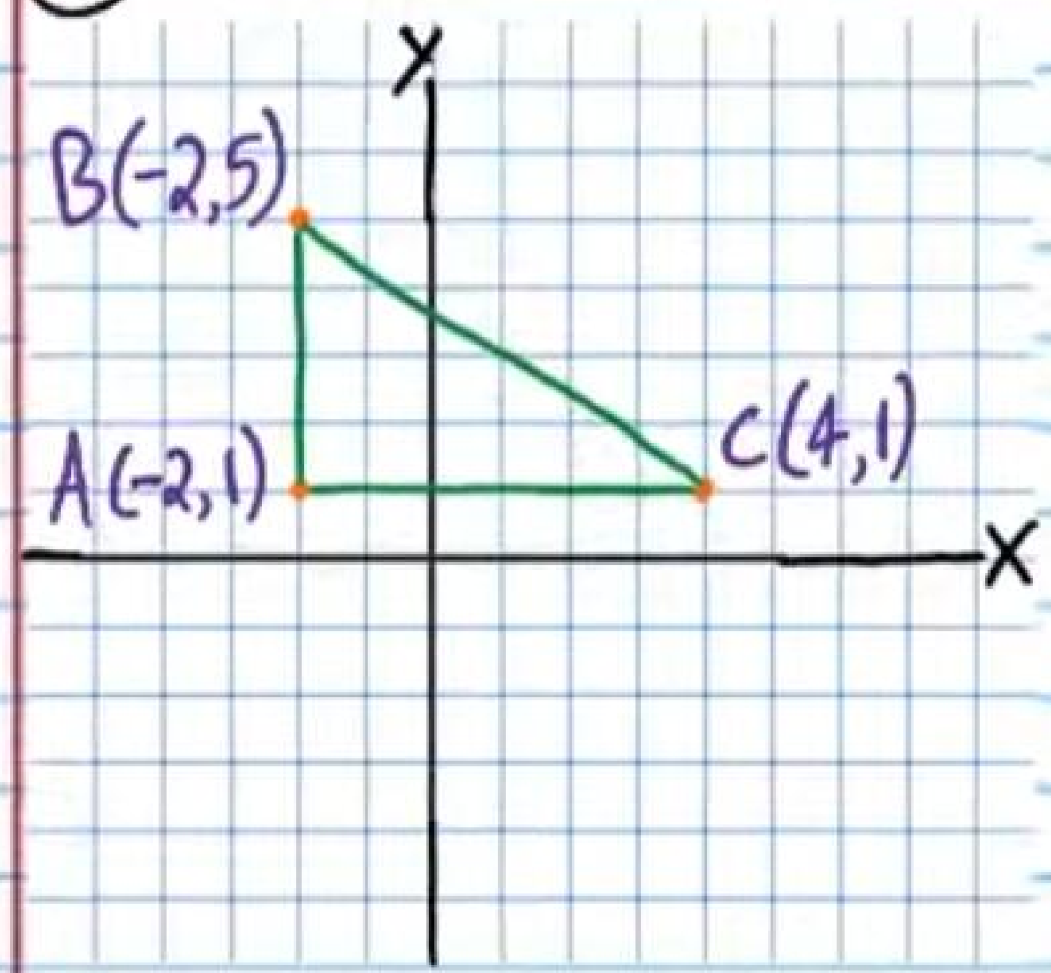
$\sqrt{HC^2} = \pm\sqrt{36}$

$HC = \pm 6$

$\boxed{HC = 6}$

iv) $BC = 6 + 6 = \boxed{12}$

(24) Find the coordinates of the orthocenter of the triangle below.



Line containing Altitude 1

$$x = -2$$

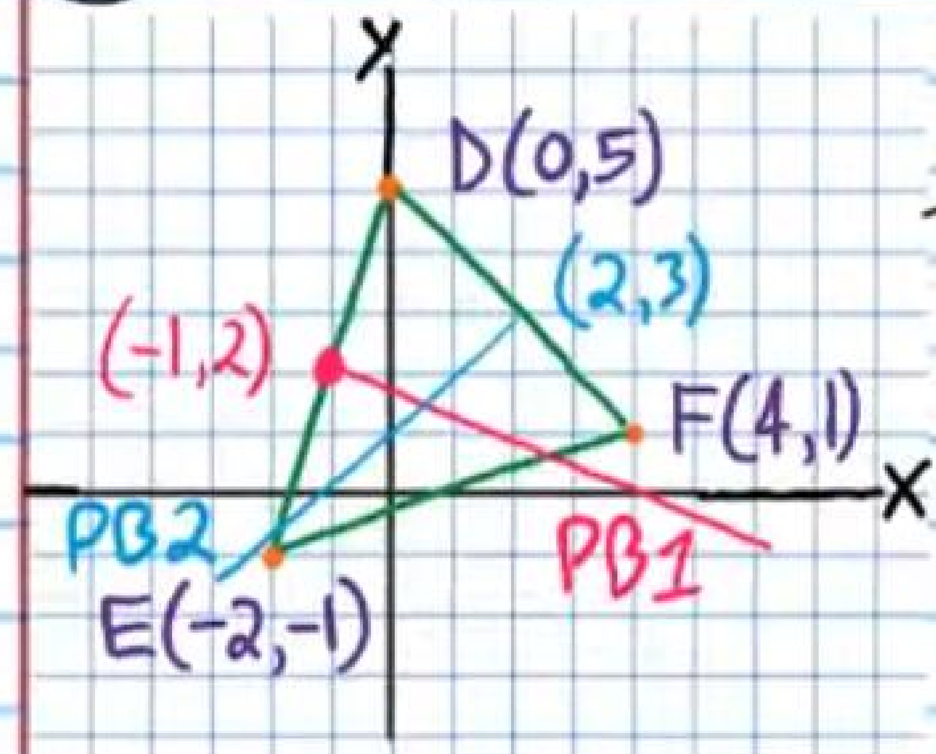
Line containing Altitude 2

$$y = 1$$

Point of Intersection

$$\boxed{(-2,1)}$$

(25) Find the coordinates of the circumcenter of the triangle below.



Line Containing PB 1

a) $M_{DE}\left(\frac{0+(-2)}{2}, \frac{5+(-1)}{2}\right) = M_{DE}(-1,2)$

b) $m_{DE} = \frac{5-(-1)}{0-(-2)} = \frac{6}{2} = 3$

$m_{\perp} = -\frac{1}{3}$

$$y - y_p = m(x - x_p)$$

$$y - 2 = -\frac{1}{3}(x-(-1))$$

$$y - 2 = -\frac{1}{3}(x+1)$$

$$y - 2 = -\frac{1}{3}x - \frac{1}{3}$$

$$y = -\frac{1}{3}x + \frac{5}{3}$$

Line Containing PB 2

$M_{DF}\left(\frac{0+4}{2}, \frac{5+1}{2}\right) = M_{DF}(2,3)$

$m_{DF} = \frac{5-1}{0-4} = \frac{4}{-4} = -1$

$m_{\perp} = 1$

$$y - 3 = 1(x-2)$$

$$y - 3 = x - 2$$

$$y = x + 1$$

Intersection Point

$$\begin{cases} y = x+1 \\ y = -\frac{1}{3}x + \frac{5}{3} \end{cases}$$

$$-\frac{1}{3}x + \frac{5}{3} = x + 1$$

$$-\frac{x}{3} + \frac{5}{3} = \frac{3x}{3} + \frac{3}{3}$$

$$\frac{-x+5}{3} = \frac{3x+3}{3}$$

$$-x + 5 = 3x + 3$$

$$2 = 4x$$

$$\frac{1}{2} = x$$

$$y = \frac{1}{2} + 1$$

$$y = \frac{3}{2}$$

$$\boxed{\left(\frac{1}{2}, \frac{3}{2}\right)}$$